

Letters to the Editor

Clinical Case Notes

Retinoblastoma, microphthalmia and the chromosome 13q deletion syndrome

ABSTRACT

A 10-month-old boy, known to have the 13q deletion syndrome, presented with leukocoria in a microphthalmic right eye. He was found to have bilateral retinoblastoma requiring enucleation of the right eye and laser, cryotherapy and external beam radiation to the left eye. This case demonstrates that retinoblastoma may be present in a microphthalmic eye and that more than one imaging modality should be used in the evaluation of suspected retinoblastoma, as calcification may be absent on the computed tomography scan but present on B-mode ultrasound. Children with a known 13q deletion require immediate and serial ophthalmological examinations.

Key words: chromosome 13q deletion syndrome, microphthalmia, retinoblastoma.

CASE REPORT

At birth, a male infant (birth weight 2365 g, 39 weeks gestation) was noted to have dysmorphic features and ocular anomalies. After further assessment he was diagnosed with the chromosome 13q deletion syndrome with the entire terminal portion of the chromosome from 13q14 distally involved. He had multiple congenital anomalies including hypotonia, facial dysmorphism, thumb and toe anomalies, cardiac malformations and bilateral congenital hip dislocations. The brain malformations consisted of agenesis of the corpus callosum, posterior fossa cysts, occipital meningocele and the Dandy–Walker malformation. There was no family history of retinoblastoma or coloboma and the parents were not consanguineous.

Ophthalmic examination on day one showed the following eye abnormalities: downslanting of the palpebral fissures, hypertelorism, bilateral posterior embryotoxon, a small right eye with an iris and retinochoroidal coloboma and left correctopia.

The patient was unable to attend ophthalmic appointments until 10 months of age when leukocoria was detected in the right eye by the ophthalmologist. There was no visual response in the right eye and the pupil did not react to light but the left eye could fix and follow light. In addition to the ophthalmic features previously found, the corneal diameter measured 8.5 mm on the right and 10 mm on the left. There was a yellow-white tumour filling a major part of the vitreous in the right eye.

As this infant was genetically at increased risk for retinoblastoma, this was the suspected diagnosis. A computed tomography (CT) scan of the right eye showed a hyperdense mass in the vitreous without calcification (Fig. 1a). B-mode ultrasound showed a large mass arising from the posterior pole (Fig. 1b). The mass was generally of medium reflectivity with a few small foci of high internal reflectivity and adjacent acoustic shadowing suggestive of calcification. The right eye was enucleated. Histological analysis confirmed a well-differentiated retinoblastoma with no invasion of the optic nerve (Fig. 2). Within the tumour there was no calcification. The multiple ocular anomalies seen clinically were confirmed microscopically.

During enucleation of the right eye, the left eye was noted to have a tumour at the equator measuring 1 mm in elevation and 3 mm in diameter and this was treated with laser and cryotherapy. At 2 years of age there was a recurrence of the tumour in the left eye and it then measured 4 mm in diameter with vitreous extension (Reese–Ellsworth classification 5B). Further laser and cryotherapy was undertaken but the tumour failed to respond so he underwent external beam radiation therapy. Chemotherapy was not considered a treatment option by the family because of his multiple anomalies and poor general health.

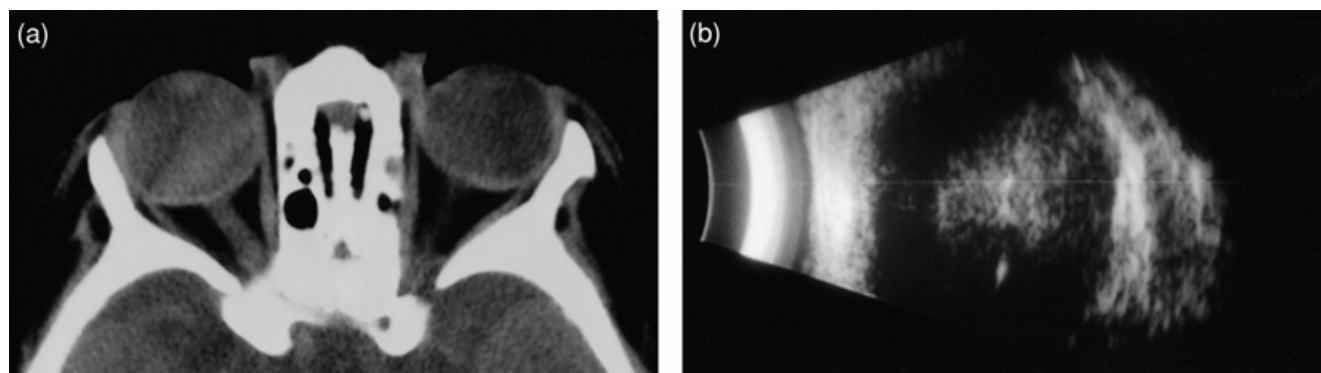


Figure 1. Right eye at 10 months of age. (a) Axial computed tomography scan showing hyperdense mass in the vitreous. (b) B-mode ultrasound of the mass, showing medium reflectivity with a few small foci of high internal reflectivity and adjacent acoustic shadowing suggestive of calcification.

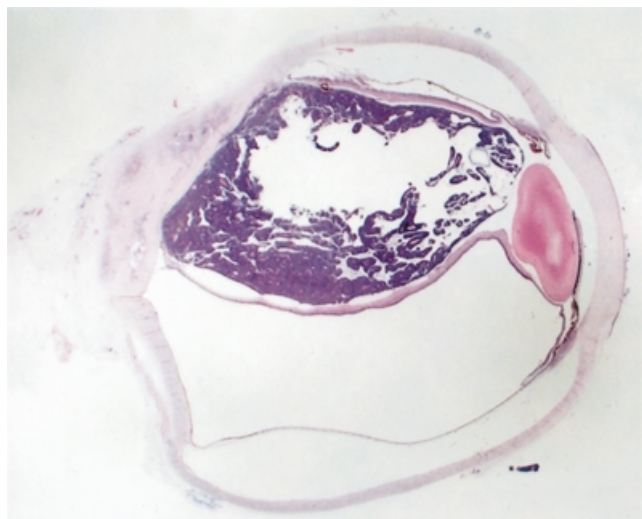


Figure 2. Histological section of the enucleated right eye showing a well-differentiated retinoblastoma without calcification.

The child is now 7 years of age and has a radiotherapy-induced neurotrophic keratitis with corneal opacification that precludes viewing of the fundus. B-mode ultrasound shows no retinal mass or calcification in the left eye. He has profound growth and mental retardation (Fig. 3).

DISCUSSION

This unusual case demonstrates that retinoblastoma can occur in a microphthalmic eye and in eyes with other anomalies, particularly the chromosome 13q deletion syndrome. It emphasizes the importance of using multiple imaging modalities in assessing children with leukocoria, even in the presence of microphthalmia, as calcification on CT scanning may be absent but is usually detected with a careful ultrasound examination.

Classic teaching holds that retinoblastoma occurs in normal sized eyes. Leukocoria presenting with microphthalmia is more likely to represent persistent hyperplastic primary vitreous or a congenital ocular anomaly. Although extremely rare, the simultaneous occurrence of retinoblastoma and microphthalmia does exist and Axelsen was the first to report a case demonstrating that one cannot be sure of the absence of tumours in small eyes.¹ Other reports now exist.^{2–5} So the classic teaching does not apply when a chromosomal deletion is present. Microphthalmia is also an extremely rare presenting feature of retinoblastoma. In a retrospective review of the presenting signs of 1265 patients with retinoblastoma, only 4 (0.3%) presented because of microphthalmia.⁶ In addition, both retinoblastoma and persistent hyperplastic primary vitreous can occur simultaneously in microphthalmic and normal sized eyes, albeit rarely.^{2,7}

It is also rare for a retinoblastoma to occur in eyes with congenital anomalies. Again, the main exception to this clinical dictum is the chromosome 13q deletion syndrome, which is characterized by intellectual and growth retardation, structural malformations of the brain and other organ systems, facial dysmorphism and a predisposition to malignancies.^{8–11} The ocular features of the chromosome 13q deletion syndrome are microphthalmos, hypertelorism and colobomata.¹² All of these ocular features were present in this



Figure 3. The patient showing profound growth retardation and dysmorphic features.

infant. Retinoblastoma occurs in about 80% of cases.⁸ Overall, 3–6% of patients with retinoblastoma demonstrate a karyotypically visible deletion of the chromosome region 13q14.¹⁰

One or more imaging studies may be required in the assessment of leukocoria, especially when the fundus of the child cannot be adequately visualized. On CT scan, evidence of calcification is shown in more than 80% of retinoblastomas.¹³ Tumours large enough to opacify the media will generally show calcium on the CT scan.¹³ Therefore, cases with advanced retinoblastoma and no evidence for calcification on CT scan, such as this one, are rare.¹⁴

Calcification is found in approximately 95% of histologically examined retinoblastomas.¹⁵ In this case, a careful B-mode ultrasound performed to specifically detect calcification did detect some high reflectivity foci even though calcification was absent on the CT and two separate pathological examinations. Although CT is more sensitive than conventional radiology in detecting intra-ocular calcium, its relative sensitivity as compared to diagnostic ocular ultrasound has not been determined.¹⁴ The absence of calcification on histopathology in this case is therefore most likely explained by its true absence or by being missed on serial sectioning.

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REFERENCES

1. Axelsen I. Retinoblastoma in a microphthalmic eye. *Ophthalmologica* 1978; **176**: 24–6.
2. Irvine AR, Albert DM, Sang DN. Retinal neoplasia and dysplasia. II. Retinoblastoma occurring with persistence and hyperplasia of the primary vitreous. *Invest Ophthalmol* 1977; **16**: 403–7.
3. Mauger TF, Makley TA, Davidorf FH, Rogers GL. Retinoblastoma, microphthalmia, coloboma, and neuroepithelioma of the pineal body. *Ann Ophthalmol* 1992; **24**: 290–94.
4. Lee JS, Shin YK, Geun J, Oum BS. Retinoblastoma which developed in microphthalmia. *Acta Ophthalmol* 1997; **75**: 730–31.
5. Hadjistilianou T, Mastrangelo D, Mazzotta C, De Francesco S, Capretti C, Lore C. 13q deletion syndrome associated with retinoblastoma and clinical anophthalmos of the opposite eye. *Med Pediatr Oncol* 2002; **38**: 293–4.
6. Abramson DH, Frank CM, Susman M, Whalen MP, Dunkel IJ, Boyd NW. Presenting signs of retinoblastoma. *J Pediatr* 1998; **132**: 505–8.
7. Liang JC, Augsburger JJ, Shields JA. Diffuse infiltrating retinoblastoma associated with persistent primary vitreous. *J Pediatr Ophthalmol Strabismus* 1985; **22**: 31–3.
8. Jones KL. Deletion 13q syndrome. In: Jones KL, ed. *Smith's Recognizable Pattern of Human Malformation*. Philadelphia: WB Saunders, 1997; 60.
9. Baud O, Cormier-Daire V, Lyonnet S, Desjardins L, Turleau C, Doz F. Dysmorphic phenotype and neurological impairment in 22 retinoblastoma patients with constitutional cytogenetic 13q deletion. *Clin Genet* 1999; **55**: 478–82.
10. Bojinova RI, Schorderet DF, Addor MC *et al.* Further delineation of the facial 13q14 deletion syndrome in 13 retinoblastoma patients. *Ophthalmic Genet* 2001; **22**: 11–18.
11. Laquis SJ, Rodriguez-Galindo C, Wilson MW, Fleming JC, Haik BG. Retinoblastoma in a patient with an X; 13 translocation and facial abnormalities consistent with 13q syndrome. *Am J Ophthalmol* 2002; **133**: 285–7.
12. Ganesh A, Kenue RK, Mitra S. Retinoblastoma and the 13q deletion syndrome. *J Pediatr Ophthalmol Strabismus* 2001; **38**: 247–50.
13. Hedges TR, Rossi-Mucelli R, Char DH, Newton TH. Computed tomographic demonstration of ocular calcification: correlations with clinical and pathologic findings. *Neuroradiology* 1982; **23**: 15–21.
14. Katz NNK, Margo CE, Dorwart RH. Computed tomography with histopathologic correlation in children with leukocoria. *J Pediatr Ophthalmol Strabismus* 1984; **21**: 50–56.
15. Bullock JD, Campbell RJ, Waller RR. Calcification in retinoblastoma. *Invest Ophthalmol Vis Sci* 1977; **16**: 252–5.

Malignant melanoma in eviscerated eyeball

ABSTRACT

A 65-year-old woman presented with malignant melanoma of the right socket following evisceration of her right eye for ocular trauma 8 years previously. Computed tomography demonstrated a solid mass adjacent to the residual eviscerated eyeball without extra-orbital extension. The tumour was removed by total exenteration of the orbit. The clinical and radiological features of this condition, and a review of the literature are presented.

Key words: evisceration, exenteration, malignant melanoma, ocular trauma.

CASE REPORT

A 65-year-old woman presented with progressive swelling in her right socket for 2 months, along with deep orbital pain referred to the frontal region. The history revealed that she had undergone evisceration of her right eye for ocular trauma 8 years previously. There was no history of weight loss or previous history of systemic malignancy.

The left visual acuity was 6/18. On examination there was oedema of the right upper and lower eyelids with an elevated, irregular brownish black mass visible within the socket (Fig. 1). The mass was firm to palpation and appeared extensively vascularized on slit-lamp examination. The mass was partly adherent to the underlying orbital bones and no identifiable ocular structures were seen.

Systemic and cutaneous examination did not reveal any evidence of primary tumour. Liver function tests, haematological investigation, chest X-ray and mammography were all normal.

Computed tomography (CT) demonstrated a large tumour mass in the right orbital cavity adjacent to remains of the eviscerated globe. It extended more on the lateral side of the orbit (Fig. 2a). The optic nerve was seen to be pushed laterally (Fig. 2b). Definitive enhancement by contrast media showed no intracranial extension of the tumour.

Biopsy showed clumps of closely packed large oval cells with eosinophilic cytoplasm suggestive of malignant melanoma.

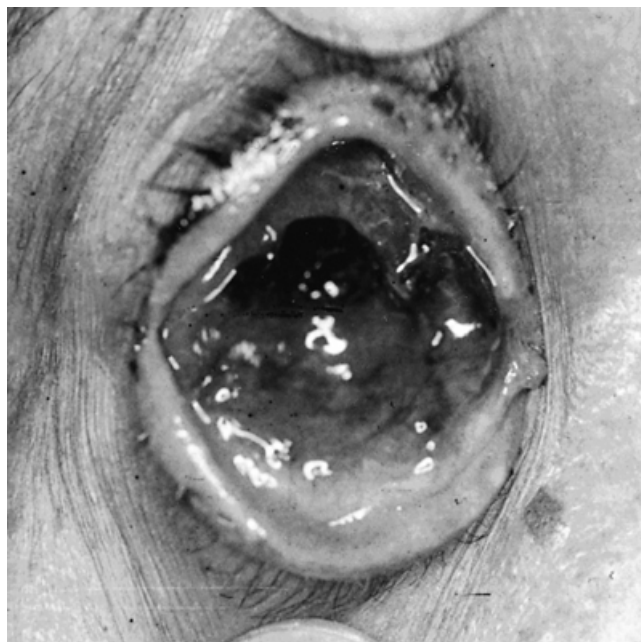


Figure 1. Clinical appearance at presentation showing encapsulated tumour mass in right orbit.

Total exenteration of the right orbit was performed. There was erosion and destruction of the periosteum and the mass was partially adherent to the underlying bone. The mass extended towards the apex of the orbit and inferolateral wall. Postoperative follow up showed healing with granulation tissue. Histopathological examination of the tumour showed epithelioid cell melanoma with clumps of closely packed large, oval cells with eosinophilic cytoplasm. The cells varied in size with hyperchromatic pleomorphic nuclei.

No recurrence was seen in 6 months of follow up. A repeat CT scan excluded tumour recurrence.

DISCUSSION

Malignant melanoma is one of the commonest primary intraocular tumours in adults. Recurrences in the orbit following enucleation have been reported.¹ Metastatic spread many years after enucleation has been reported as a delayed presentation of malignant melanoma.²

Although malignant melanomas have been discovered in eyes after evisceration,^{3,4} this is a rare occurrence. In our present case, evisceration was performed for penetrating ocular injury in an eye with no visual prognosis. Penetrating trauma as an aetiology for malignant melanoma was found in 11% of all melanoma cases, according to Fuchs in 1882, and none of these cases had undergone evisceration or enucleation.⁵ The occurrence of malignant melanoma in our present case in an eviscerated eye is due to the presence of residual uveal pigment in an eviscerated eyeball.⁶

Evisceration should be approached with caution in eyes with opaque media.⁷ We recommend careful examination of eviscerated contents of traumatized eyes with opaque media, and long-term follow up of these patients to exclude any associated presence of malignant melanoma of the uveal tract.

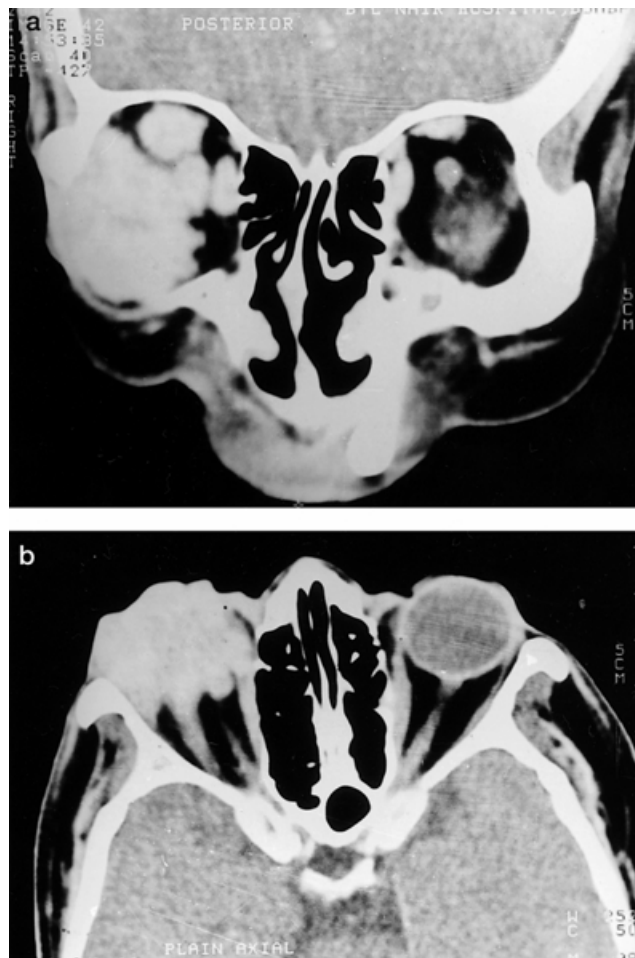


Figure 2. Computed tomography scan at presentation. (a) Coronal section showing density of mass in lateral portion of right orbital cavity. (b) Tumour mass filling anterior portion of right orbital cavity displacing optic nerve laterally.

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REFERENCES

1. Allen JC, Jaeschke WH. Recurrence of malignant melanoma in an orbit after 28 years. *Arch Ophthalmol* 1966; 76: 79–81.
2. Hogan MJ, Zimmerman LE. *Ophthalmic Pathology*, 2nd edn. Philadelphia: WB Saunders, 1962; 429.
3. Star HJ, Zimmerman LE. External extension and orbital recurrence of malignant melanoma of the choroid and ciliary body. *Int Ophthalmol Clin* 1962; 2: 369–85.
4. Smith B, Soll DB. Malignant melanoma of choroid: report of a case discovered after evisceration. *Am J Ophthalmol* 1960; 50: 330–33.

5. Duke-Elder S, Perkins ES. Cysts and tumours of the uveal tract. In: Duke-Elder S, ed. *System of Ophthalmology*, Vol. IX, 1st edn. London: Henry Kimpton, 1977; 846.
6. Kita Y, Yabe H, Shikishima K, Takahashi K. A case of malignant melanoma occurring 63 years after evisceration. *Acta Soc Ophthalmol Jpn* 2001; **105**: 52–7.
7. MacDonald R, Edwards WC. Melanoma of the orbit: an interesting case following evisceration. *Surv Ophthalmol* 1967; **12**: 253–7.

Tamoxifen optic neuropathy

ABSTRACT

A 68-year-old woman with metastatic breast cancer presented with slowly progressive visual loss in each eye. Electrophysiology confirmed bilateral optic neuropathies. Investigations to determine the cause of the vision loss were negative. Cessation of tamoxifen resulted in a dramatic improvement in acuity.

Key words: breast cancer, optic neuropathy, side-effect, tamoxifen.

INTRODUCTION

Tamoxifen, a non-steroid anti-oestrogen agent, is the chemotherapeutic agent of choice for metastatic breast cancer in postmenopausal women. Ophthalmic side-effects, including retinopathy and keratopathy, are well documented. Three cases of coexistent bilateral optic neuritis and tamoxifen administration have been described.^{1–3} We report the first case of slowly progressive vision loss in a patient on tamoxifen suggestive of a toxic neuropathy. Our patient showed dramatic improvement on cessation of tamoxifen.

CASE REPORT

A 68-year-old woman presented with progressive visual loss over 4 months. Eighteen months previously she had undergone a simple mastectomy for breast carcinoma. Metastatic disease affecting the chest wall was diagnosed 4 months later and was treated with radiotherapy. Tamoxifen 20 mg/day was commenced.

On examination corrected visual acuity was 6/36 in the left eye and 6/18 in the right eye. A small left afferent pupillary defect was present. Red colour desaturation was detected in the left eye on testing with the top of a tropicamide bottle. There was no proptosis and the eye movements were normal. Slit-lamp and dilated fundus examination were entirely normal.

Goldmann fields demonstrated generalized constriction of the left field with a paracentral scotoma and mild constriction on the right (Fig. 1). Full blood count, erythrocyte sedimentation rate, folate and Vitamin B12 levels performed on initial presentation were normal.

Magnetic resonance imaging was performed 1 week later. This was undertaken on a 1.5 Tesla machine and consisted of axial T2 and coronal T1 imaging through the brain with axial and coronal T1 weighted scans after gadolinium administration, supplemented by high resolution axial sequences through the orbits. The anterior visual pathways were well demonstrated and showed no compressive lesion or pathological enhancement.

Electrophysiological tests were performed and showed a normal electroretinogram but markedly reduced visual evoked cortical potentials bilaterally, almost extinguished on the left.

Further investigations to elucidate the cause of the bilateral optic neuropathy included syphilis serology, serum electrophoresis, Leber's mutational analysis, antinuclear antibodies and antineutrophil cytoplasmic antibodies. All these tests were normal.

A lumbar puncture was performed. Cerebrospinal fluid protein 0.3 g/L (0–0.4) and glucose 3.5 mmol/L (70% of blood glucose) were within normal limits. No white or red cells were seen. No oligoclonal bands were present. Cytology did not identify malignant cells.

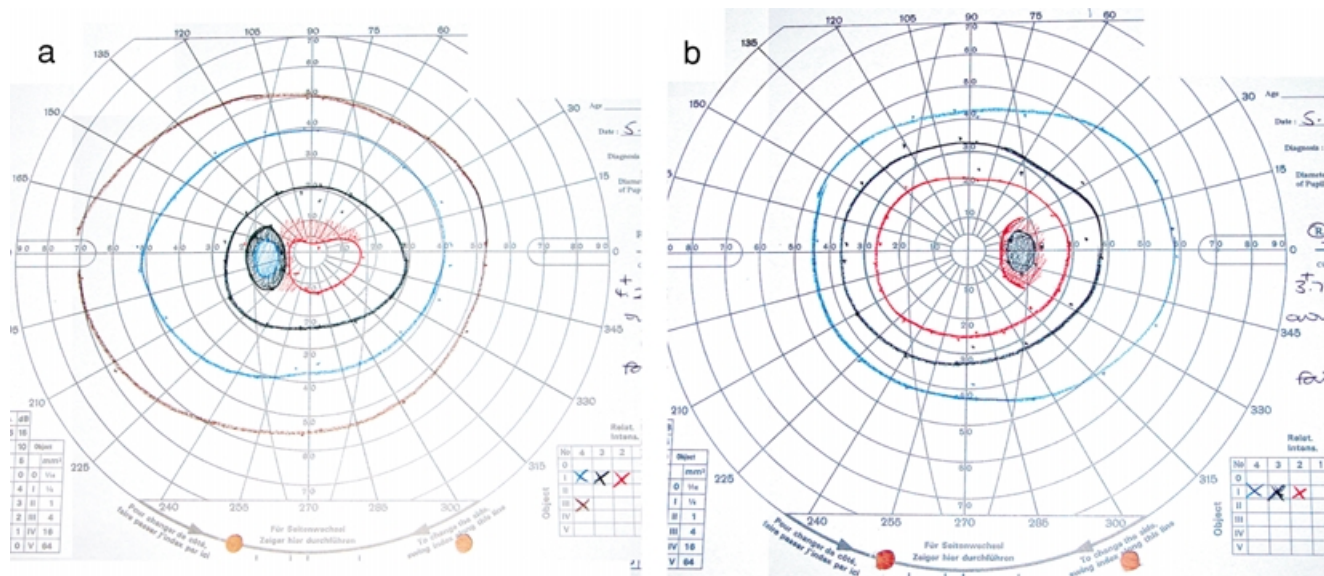


Figure 1. Goldmann visual fields of (a) left eye and (b) right eye at presentation, showing a left centrocaecal scotoma.

Table 1. Criteria for diagnosis of toxic optic neuropathies⁴

History of drug or toxin exposure
Gradual onset of bilateral, symmetrical impaired visual acuity and colour vision
Painless visual loss
Visual field defects, typically central or cecentral scotomata
Slowly progressive course
Optic nerve may appear normal until late, but optic atrophy (often temporal pallor) develops eventually; rarely initial disc swelling

Serum and cerebrospinal fluid samples were negative for paraneoplastic antineuronal antibodies (anti-Hu, anti-Yo, and anti-Ri).

On review 2 months after presentation the patient's vision had deteriorated to count fingers left and 6/36 right. A left optic nerve and nerve sheath biopsy was undertaken to exclude tumour infiltration. Tamoxifen was ceased and substituted with anastrozole. High-dose steroid therapy was commenced postoperatively. The biopsy demonstrated fragments of normal optic nerve and sheath with no specific pathological features.

Follow up 1 month later showed no change in acuity and severely constricted fields bilaterally. The left optic disc was pale. Steroids were tailed off.

The next review, 5 months after presentation, showed a dramatic improvement in the right visual acuity to 6/18, but the left visual function was unchanged. The left optic disc was almost totally atrophic and the right disc showed some temporal pallor. The right visual field improved to that seen on presentation.

Visual acuity in the right eye recovered to 6/9⁺, 18 months after onset of symptoms, but no further improvement in colour vision was demonstrated.

DISCUSSION

This patient presented with progressive visual loss that would fit the diagnostic criteria for a toxic optic neuropathy (Table 1). As this cannot be established with absolute certainty without re-exposure to Tamoxifen, other causes including nutritional deficiencies, compressive lesions, infective and inflammatory neuropathies should be ruled out.

In the setting of known metastatic malignancy both paraneoplastic and autoimmune optic neuropathies may be causal. Treatment of the specific cancer in patients with paraneoplastic optic neuropathy and the use of high dose steroids in autoimmune optic neuropathy can result in significant visual improvement. Against these diagnoses were the absence of antineuronal antibodies, normal antinuclear antibodies and the absence of enhancement of the optic nerves on magnetic resonance imaging.

The incidence of retinopathy from tamoxifen may be as high as 5%.³ This toxicity has been reversible in the majority of cases. It is likely that tamoxifen toxic optic neuropathy is a rare side-effect of this treatment but it may be an under-diagnosed entity if it coexists in patients with retinopathy and reduced vision.

Toxic optic neuropathy should be considered in patients who develop visual impairment while receiving tamoxifen therapy. Other causes of bilateral optic neuropathy should be ruled out, but prompt cessation of tamoxifen may result in visual recovery.

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REFERENCES

1. Pugesgaard T, Von Eyben FE. Bilateral optic neuritis evolved during tamoxifen treatment. *Cancer* 1986; **58**: 383–6.
2. Ashford AR, Donev I, Tiwari RP, Garrett TJ. Reversible ocular toxicity related to tamoxifen therapy. *Cancer* 1988; **61**: 33–5.
3. Nouredin BN, Seoud M, Bashshur Z, Salem Z, Shamseddin A, Khalil A. Ocular toxicity in low-dose tamoxifen: a prospective study. *Eye* 1999; **13**: 729–33.
4. Brazis PW, Lee AG. Neuro-ophthalmic problems caused by medications. *Focal Points* 1998; **16**: 1–14.

Propecia-associated bilateral cataract

ABSTRACT

A 43-year-old man presented suffering from decreasing vision in both eyes for 3 months. The patient's visual acuity was 6/20 (non-corrected) in the right eye and 6/10 (−1.75/−1.00 × 91) in the left. Ocular examination of both eyes revealed anterior subcapsular opacities of both lenses with the right eye being more severe than the left. He had been taking finasteride (Propecia; Merck, Sharp and Dohme) at 1 mg/day for 3 years to treat early stage of androgenic alopecia. It was highly suspected that finasteride was associated with the anterior subcapsular opacity on the lens, and the patient therefore discontinued use of finasteride. He underwent uneventful cataract extraction surgery and intraocular lens implantation of the right eye. One month after cataract surgery in the right eye, the best-corrected visual acuity was right 6/6 (−1.25) and left 6/10 (−2.00/−0.50 × 100). To the best of the authors' knowledge, this is the first reported case of Propecia-associated cataract.

Key words: androgenic alopecia, cataract, finasteride, Propecia, subcapsular cataract.

INTRODUCTION

On milligram per day administration of oral finasteride (Propecia; Merck, Sharp and Dohme, Whitehouse Station, NJ, USA) has been widely used as an effective treatment for male androgenic alopecia.¹ It has been extensively investigated in clinical trials, all of which have reported excellent tolerability.^{2–5} The most commonly reported side-effects of the drug are decreased libido, erectile dysfunction, ejaculatory dysfunction (3–5%), and thrombocytopenia.⁶

The case is presented herein of a 43-year-old man who developed bilateral cataracts following use of Propecia for 3 years to treat male pattern baldness. A wide variety of cataracts have been reported in the literature as being drug-induced; however, to our knowledge, finasteride-associated cataract has not been one of the previously reported conditions.

CASE REPORT

At initial presentation to our clinic, a 43-year-old man complained of increasing glare and poor vision for 3 months. His visual acuity was 6/20 (non-corrected) in the right eye and 6/10 (−1.75/−1.00 × 91) in the left. Ocular examination revealed bilateral anterior subcapsular lens opacities, with the right eye being more



Figure 1. Anterior subcapsular opacity on the right lens.

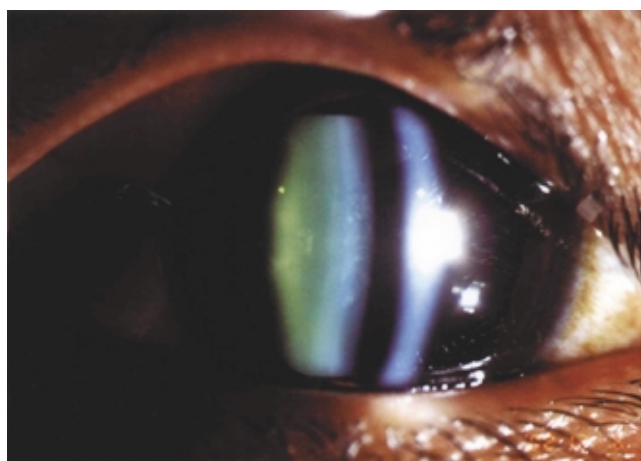


Figure 2. Mild anterior subcapsular opacity on the left lens.

affected than the left (Figs 1–3). The rest of his ocular examination was normal.

He had previously been healthy and regularly took only Propecia (finasteride) at 1 mg/day for 3 years due to early stage androgenic alopecia. He was on no other medications. He had no other significant past history, including no history of trauma or congenital cataracts. The alopecia had improved with the medication and no other side-effects were noted. Laboratory investigations were essentially normal, with normal renal and liver function.

Therefore, finasteride was suspected as a possible cause of the anterior subcapsular cataracts and use of it was discontinued. The patient subsequently underwent cataract extraction and intraocular lens implantation in the right eye. One month after cataract surgery, his best corrected visual acuity was right 6/6 (−1.25) and left 6/10 (−2.00/−0.50 × 100).

DISCUSSION

Finasteride is a type II 5- α -reductase inhibitor that inhibits conversion of testosterone to dihydrotestosterone (DHT). The drug is effective in the treatment of androgenetic alopecia where follicle miniaturization occurs as a consequence of androgen

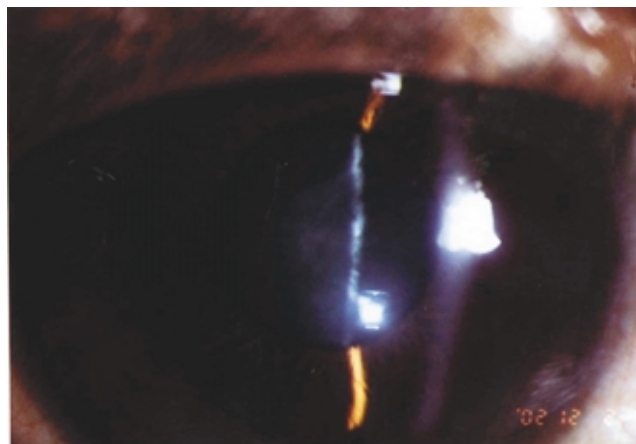


Figure 3. Narrow slit beam on the right lens.

(mainly DHT) action on the hair follicle. Administration of finasteride 1 mg (Propecia) significantly lowers serum, prostate and scalp DHT levels and products, with a slight increase in testosterone levels, which, however, remain within physiological limits.⁷ Our patient did not have any side-effects at all. However, he suffered from decreasing vision due to anterior subcapsular cataract formation.

Drug-induced lens changes have been noted in corticosteroid, phenothiazine, miotics and amiodarone use. Long-term use of corticosteroids may cause posterior subcapsular cataracts. Their occurrence is related to dose and duration of treatment, and individual susceptibility to corticosteroid-induced posterior subcapsular cataracts appears to vary. Some steroid-induced posterior subcapsular cataract in children may be reversible with cessation of the drug. Phenothiazines, a major group of psychotropic medication, can cause pigmented deposits in the anterior lens epithelium in an axial configuration. These deposits appear to be affected by both dose and duration. The visual changes associated with phenothiazine use are generally insignificant. Anticholinesterases such as echothiophate iodide and demecarium bromide can cause cataracts. Usually, these cataracts appear as small vacuoles within and posterior to the anterior lens capsule and epithelium. The cataract may progress to posterior cortical and nuclear lens changes as well. Amiodarone has also been reported to cause stellate anterior axial pigment deposition.⁸

Propecia-associated cataract has not previously been reported. Our patient was noted to have bilateral anterior subcapsular cataract following finasteride use for 3 years. The right lens opacity appeared more severe than the left. After cataract extraction, the visual acuity in the right eye improved from 6/20 (non-corrected) to 6/6 (with correction). There was no change in the appearance of the left lens following discontinuation of the medication. The patient had no other significant past history and was on no other medications. He also had no family history of early onset cataract formation or hereditary disease.

The clinical appearance of the cataracts in both eyes was compatible with the clinical appearance of cataracts caused by other drugs, such as amiodarone and phenothiazine. Patients using finasteride for benign prostate hyperplasia are usually old and the ophthalmologist might not consider the possibility that anterior subcapsular lens opacity is due to finasteride treatment as old age is often associated with lens changes. Younger patients with male

androgenic alopecia dislike long-term use of finasteride because of commonly reported side-effects related to sexual dysfunction. The finasteride-related anterior subcapsular opacities might consist of deposits of the drug, such as occurs in phenothiazine- and amiodarone-treated patients. Furthermore, phenothiazine and amiodarone deposits are also affected by drug dose, duration of treatment, and individual susceptibility. For the anterior subcapsular opacities to form in our case, finasteride of more than 3 years might have been needed. However, we were not able to obtain lens material for further analysis.

These cataracts might be associated with alopecia because they are both ectodermal disorders. Appell *et al.* reported the study of a kindred with numerous ectodermal abnormalities, including childhood-onset alopecia, widespread keratosis pilaris and premature cataracts.⁹ Disorders such as Werner's syndrome and alopecia areata were also noted with alopecia or hypotrichosis and cataracts.⁹ Alopecia areata patients have asymptomatic, isolated areas of hair loss usually of the scalp, although any area of the body may be involved. Ocular alterations with development of cataract in alopecia areata had been reported with dermatological findings such as long-standing atopic dermatitis.¹⁰ However, none of the dermatological findings were found in our patient or his family.

Our patient's vision (previously 6/6 in both eyes) became progressively poorer after using Propecia for 3 years. Propecia might be associated with the formation of anterior subcapsular lens opacities. Although finasteride is a commonly used drug with an excellent side-effects profile, it can be associated with the development of cataracts. Clinicians should therefore be aware of this and report suspected cases to the appropriate regulatory bodies.

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REFERENCES

1. Kaufman KD, Olsen E, Whiting D *et al.* Finasteride 1 mg/day in the treatment of men with androgenetic alopecia. *J Am Acad Dermatol* 1998; **39**: 578–89.
2. Stoner E, Gregg H, Otterbein JJ *et al.* Administration of MK-906, a 4-azasteroid compound, results in marked suppression of serum dihydrotestosterone in healthy men. *Clin Res* 1987; **35**: 402A.
3. Gormley GJ, Stoner E, Bruskewitz RC *et al.* The effect of finasteride in men with benign prostatic hyperplasia. The Finasteride Study Group. *N Engl J Med* 1992; **327**: 1185–91.
4. Andersen JT, Ekman P, Wolff H, Scandinavian BPH Study Group. Can finasteride reverse the progress of benign prostatic hyperplasia? A two-year placebo-controlled study. *Urology* 1995; **46**: 631–7.
5. Marberger MJ, on behalf of the PROWESS Study Group. Long-term effects of finasteride in patients with benign prostatic hyperplasia: a double-blind, placebo controlled, multi-centre study. *Urology* 1998; **51**: 677–85.
6. Lepor H, Williford WO, Barry MJ, Veterans' Affairs Study Group. The efficacy of terazosin, finasteride, or both in benign prostatic hyperplasia. *N Engl J Med* 1996; **335**: 533–9.

7. Tosti A, Piraccini BM, Soli M. Evaluation of sexual function in subjects taking finasteride for the treatment of androgenetic alopecia. *J Eur Acad Dermatol Venerol* 2001; **15**: 418–21.
8. Weingeist TA, Liesegang TJ, Grand MG. Lens and cataracts. In: Weingeist TA, Liesegang TJ, Grand MG, eds. *Basic Science and Clinical Course*. San Francisco: American Academy of Ophthalmology, 1999; 49–52.
9. Appell ML, Sherertz EF. A kindred with alopecia, keratosis, pilaris, cataracts, and psoriasis. *J Am Acad Dermatol* 1987; **16**: 89–95.
10. Recupero SM, Abdolrahimzadeh S, Dominicus M *et al.* Ocular alterations in alopecia areata. *Eye* 1999; **13**: 643–6.

Castleman's disease of the lacrimal gland

ABSTRACT

Castleman's disease, also known as angiofollicular lymphoid hyperplasia, is a lymphoproliferative disorder that is generally benign. The lesions most commonly originate in the lymph nodes of the mediastinum. The second documented case of Castleman's disease of the lacrimal gland is reported in an 84-year-old woman who presented with an 8-year history of a left upper lid mass. This was excised via a lateral orbitotomy, and a diagnosis of Castleman's disease was made histologically. Relevant images of the clinical findings and histology are shown.

Key words: angiofollicular lymphoid hyperplasia, Castleman's disease, lacrimal gland mass, orbital mass.

INTRODUCTION

Benjamin Castleman first described Castleman's disease in 1954.¹ It is an idiopathic lymphoproliferative disorder, which affects the lymph nodes of the mediastinum in 70% of cases.² Involvement of the lacrimal gland has been described only once before in the literature.³

CASE REPORT

An 84-year-old woman presented to her ophthalmologist for a routine check for left pseudoexfoliation glaucoma. It was noted that she had a left upper lid mass, which, on further questioning, was stated to have been present for more than 8 years. In recent months, she noted the mass had slowly enlarged. It caused slight discomfort, insufficient to require analgesics. She suffered no diplopia and no deterioration in her visual acuity. Systems enquiry was negative; specifically, she suffered no weight loss, weakness, fevers, night sweats, or shortness of breath. Her past ocular history included bilateral pseudoexfoliation (PXF), with moderate left PXF glaucoma. She had previously been treated with left argon laser trabeculoplasty, and was controlled on pilocarpine 1% q.i.d. left eye, and timolol 0.5% b.d. left eye. Her medical history included primary hypothyroidism for which she took thyroxine, asymptomatic seronegative rheumatoid arthritis and cardiomegaly.

On examination she had an S-shaped left upper lid ptosis. A firm, rubbery, and slightly tender, mobile mass was palpable



Figure 1. (a) Coronal and (b) axial computed tomography scans showing a smooth homogeneous mass arising from the left lacrimal gland. Note globe displacement. No changes to the bony orbit were apparent.

superotemporally, protruding from beneath the orbital rim. Auscultation of the left eye was negative, and the mass did not change with head posture. Hertel exophthalmometry showed a left proptosis of 4 mm, with the globe slightly displaced inferiorly and medially. Her eye movements were full. Her best-corrected visual acuities were 6/7.5 right and 6/18 left. There was no afferent pupillary defect, and the Ishihara plates were all read correctly with both eyes. There were bilateral cataracts, left greater than right, and bilateral PXF. The optic nerve heads showed asymmetry, with cup:disc ratios of 0.5 right and 0.7 left, consistent with her glaucoma. A Humphrey visual field analysis was normal in the right eye, and showed a dense left inferior arcuate scotoma. She had early changes of age-related macular degeneration bilaterally. She had no peripheral lymphadenopathy, and no hepatosplenomegaly.

A computed tomography (CT) scan of her orbits showed a well-defined, homogeneous oval mass arising from the left lacrimal gland (Fig. 1). The mass was surrounded by normal lacrimal gland tissue. There were no abnormal bony changes. The chest X-ray was normal. A full blood count and rheumatoid factor were normal and an erythrocyte sedimentation rate (ESR) was not performed.

Although the examination findings were suggestive of a lymphoproliferative lesion, a benign epithelial tumour of the lacrimal gland was considered due to the duration of the history. After

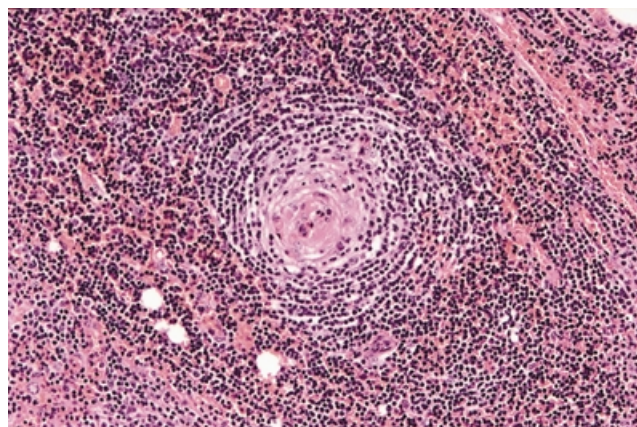


Figure 2. Histology of the lacrimal gland showing the small follicular structure with venule and perifollicular concentric arrangement of small lymphocytes. (H&E, ×200 original magnification.)

discussion with the patient, we elected to proceed with en bloc removal of the lacrimal gland and periorbita using a lateral orbitotomy approach.

PATHOLOGY

The histology (Fig. 2) showed findings consistent with the hyaline vascular variant of Castleman's disease. The mass consisted of a lymphoid proliferation that included abnormal germinal centres containing small follicles penetrated by hyalinized venules. These were often surrounded by an 'onion skin' concentric arrangement of small lymphocytes. Residual glandular structures consistent with lacrimal gland were present within the lymphoid infiltrate.

Cell marker studies performed on fresh cell suspension showed a polyclonal, predominantly B-cell population, a finding against the alternative diagnosis of low grade B-cell lymphoma.

DISCUSSION

Castleman's disease is an idiopathic lymphoproliferative disorder,¹ which may present with localized or generalized disease, with or without systemic involvement. There are two histological variants, the hyaline-vascular and plasma cell types.² Involvement of the lacrimal gland is rare.³

Whether localized disease evolves into a multicentric (or generalized) form is uncertain but some of the systemic features are shared by both forms. The hyaline vascular histological variant accounts for 80–90% of cases, is usually localized and is infrequently associated with minor haematological abnormalities. Localized disease tends to disappear with treatment. The plasma cell type involves multiple and mainly peripheral lymph nodes. It is associated with severe systemic symptoms (fever, sweats, loss of weight), and abnormal laboratory findings (anaemia, elevated ESR, hypergammaglobulinaemia, and hypoalbuminaemia). Multicentric (or generalized) disease may undergo malignant transformation. A mixed type is the least common.^{2,4}

The aetiology is unknown, but two proposed theories exist. One theory is of an exaggerated inflammatory response (infectious or non-infectious). Another suggests that it is a vascular hamartoma.²

Localized Castleman's disease is usually cured by excision.^{2,4} Corticosteroids, chemotherapy and radiotherapy, with variable responses, have also been used.⁵

Snead *et al.* reported a single case of right lacrimal fossa Castleman's disease, also of the hyaline vascular type. This patient was treated with excision biopsy, via a lateral orbitotomy, and had no recurrence on CT scan at 5 years.³ There is one reported case of multicentric Castleman's disease, with presumed (unbiopsied) Castleman's disease of the optic nerve and choroid. This patient had histologically proven Castleman's disease of the lung, although the histology was not described, and an unbiopsied, small right parietal paraventricular-enhancing lesion on a brain CT scan. The patient was treated with radiotherapy to the eye, and the associated exudative retinal detachment resolved over several months.⁵

This is only the second documented case of Castleman's disease of the lacrimal gland. Our patient presented with a primary orbital mass that we believe represents localized Castleman's disease.

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REFERENCES

1. Castleman B, Iverson L, Menendez VP. Localized mediastinal lymph node hyperplasia resembling thymoma. *Cancer* 1956; **9**: 822–30.
2. Keller AR, Hochholzer L, Castleman B. Hyaline-vascular and plasma-cell types of giant lymph node hyperplasia of the mediastinum and other locations. *Cancer* 1972; **29**: 670–83.
3. Snead MP, James JN, Snead DR, Robson DK, Rizk SN. Orbital lymphomas and Castleman's disease. *Eye* 1993; **7**: 84–8.
4. Frizzera G. Castleman's disease: more questions than answers. *Hum Pathol* 1985; **16**: 202–5.
5. Gittinger JW Jr. Ocular involvement in Castleman's disease. Response to radiotherapy. *Ophthalmology* 1989; **96**: 1646–9.

Lipoid proteinosis: a rare disorder with pathognomonic lid lesions

ABSTRACT

Lipoid proteinosis is a rare disorder seen even more rarely by ophthalmologists. The interesting clinical and histopathological features of this disease are described in a brother and sister who presented predominantly with lid lesions, without any systemic complaints. Knowledge of the typical lid lesions may help to diagnose this disorder.

Key words: lid nodules, lid papules, lipoid proteinosis, skin lesions.

INTRODUCTION

Lipoid proteinosis (Urbach–Wiethe disease) is a very rare, recessively inherited disorder characterized by infiltration of hyaline

material into the skin, oral cavity, larynx and internal organs.¹ Although it has been described in dermatology literature, there is not much information in the ophthalmic literature, probably because it usually presents in infancy with hoarseness¹ and not with an ophthalmic complaint. It is initially recognized as a dermatological and otolaryngological disorder owing to its characteristic cutaneous lesions and involvement of oral mucosa, pharynx and larynx.² However, lid lesions, consisting of beaded papules along the lid margins, are considered almost pathognomonic of the disease.^{1,3} Rarely, a patient may present first to an ophthalmologist as in the two siblings reported here. We report the clinical features, which may help ophthalmologists to diagnose the underlying disease.

CASE REPORTS

Two siblings, a boy of 14 years age and a girl of 10 years age, presented to us with a complaint of nodules on the eyelids, which were noticed by the patients 3–4 years ago. They increased in severity for about 1 year, after which they became static. There was also a history of dryness of the skin with occasional papules and repeated infections, for the last 7–8 years, more severe in the brother. This was associated with loss of the eyelashes and eyebrows. There was no history of hoarseness, light sensitivity, epilepsy, rage attacks or dental abnormalities. There was no history of the parents being consanguineous.

On examination, both siblings had rows of flesh coloured nodules along the lid margins (Fig. 1). There was a single row in the lower lid occupying more than half of the lid margin and 2–3 rows in the upper lids spanning the entire lid margin (Fig. 2). Some eyelashes in the upper lids were abnormally directed but none of them was rubbing the cornea as the misdirection was mild. The skin showed generalized dry, scaly appearance, especially over the face and neck, and there were non-inflammatory, hypo and hyper pigmented keratotic macular lesions on the arms, legs and neck. There were also scars left by previous pustular skin infections. A clinical diagnosis of lipoid proteinosis was made on the basis of the characteristic lid lesions associated with the skin lesions.

A biopsy was taken from the affected skin of the neck. Histopathological examination showed acanthosis, hyperkeratosis and parakeratosis of the skin. The reticular dermis showed mild perivascular, chronic inflammatory cell infiltrate. There was deposition of Periodic Acid-Schiff (PAS) positive, diastase resistant material in the upper dermis as well as around the blood vessels. Congo red and oil red-O stains showed no reaction. These features were consistent with a diagnosis of lipoid proteinosis.

The parents were counselled about the disease and no further intervention was advised.

DISCUSSION

Lipoid proteinosis was first described by Urbach and Wiethe in 1929.² Also called Urbach–Wiethe disease or hyalinosis cutis et mucosae, lipoid proteinosis is a recessively inherited disorder³ characterized by non-inflammatory, persistent papules on the skin and mucous membranes.^{4,5} This disease has a strong predilection for white races³ and an increased incidence is seen in South Africa and Sweden.⁶ There is no sexual predilection.³

Lipoid proteinosis presents early in life.¹ Hoarseness develops in infancy,^{1,4,5} becomes prominent within the first few years of life

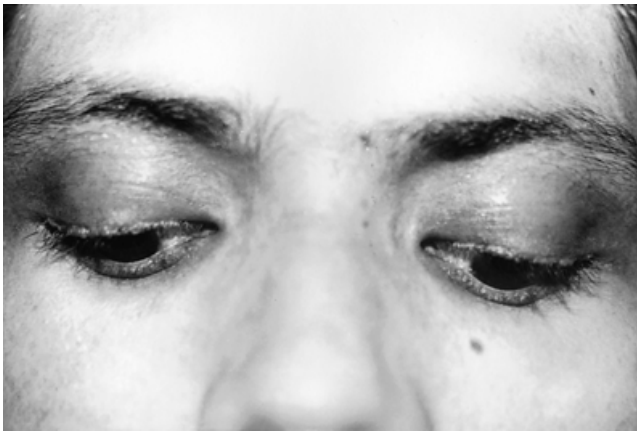


Figure 1. Clinical photograph showing the characteristic 'string of beads' appearance of the upper and lower lids in the 14-year-old boy.

and can progress to complete aphonia.¹ It is caused by infiltration of the laryngeal mucosa and is the first clinical sign.⁴ Mucosae of the pharynx, tongue and lips soon develop firm, yellow-white infiltrates.^{1,4,5} The tongue is enlarged and feels firm on palpation.^{1,4,5} Skin changes become prominent in early life with the development of yellow-brown nodules on the face and lips.¹ Scattered lesions resembling atrophic, pitted, acne scars¹ may be seen on the face as well as on non-acne prone regions of the body.⁴ Deposition of yellow material induces a marked thickening of the facial skin with deep wrinkles, which may resemble solar elastosis.⁴ Translucent, keratotic papules are seen on the elbows and knees.⁷ The skin shows increased susceptibility to trauma and infection with recurrent attacks of impetigo, often bullous in nature.⁸ Characteristic bilateral symmetric calcifications or ossifications are found in the anteromedial temporal lobes¹ in 50–75% of cases⁸ over 10 years of age.⁸ Dental abnormalities are seen in 30% cases⁴ and epilepsy may also be seen.¹

The lid lesions are described in 50% of cases.^{3,8} These lesions take the form of typical, small, flesh coloured papules seen along the margins of the upper and lower eyelid.^{1,2} The appearance of these papules is variously described as 'string of beads' or 'eyelid beading'⁹ and is also known as 'moniliform blepharosis'.⁸ Infiltration of the eyelids may induce malfunction and loss or misdirection of lashes³ causing corneal ulcers.⁴

The disease runs a chronic but benign course.³ The general prognosis is good¹ and the disease is compatible with a normal life span.⁴ However, the quality of life is seriously impaired by the disfiguring lesions and the permanent hoarseness.⁴ The diagnosis depends on the characteristic clinical features including hoarseness, thickening and stiffening of the tongue and characteristic cutaneous lesions, including the lid lesions.^{1,9} There may be unusual presentation in adults with cutaneous lesions being absent or barely evident.² A skin biopsy confirms the diagnosis.

On histopathology, epithelium shows hyperkeratosis and irregular acanthosis.^{1,5} Dermis is considerably thickened with extensive deposits of extracellular, homogenous, hyaline material⁹ seen in the upper half.^{1,5} The hyaline material is at first deposited along the course of capillaries,² in arrector pili muscles and concentrically around sweat coils.^{1,5,10} In older lesions, broad bands of hyaline material are found in the dermis in a vertical orientation.¹ This



Figure 2. Close-up photograph of the lid nodules of the 10-year-old girl showing the characteristic flesh coloured nodules. There is only one misdirected lash in the lower lid.

hyaline material stains strongly with PAS stain.^{1,5,9} Some areas also show positive staining with oil red-O.² Immediately surrounding the blood vessels, there is reduplication of basal laminae² in an 'onion skin' arrangement.^{1,2,6} It is suggested that overproduction of basement membrane collagens (Type IV and V) by the endothelial cells of blood vessels and an underproduction of fibrous collagens (Type I and II)² by the fibroblasts associated with an increased synthesis of non-collagenous structural glycoprotein by the fibroblasts is the underlying disorder.¹⁰ The disease must be differentiated from erythropoietic porphyria.⁵ In the latter, the lesions are found mainly in the sun-exposed areas^{4,5} and the mucous membranes are not involved; the hyaline is not so extensively deposited and is never found around sweat coils.⁵ Also, there is no light sensitivity in lipid proteinosis.⁴ Post-mortem studies have shown that lipid proteinosis is a generalized disorder with irregular and widespread distribution in organs.²

Lipoid proteinosis is a rare disorder not usually seen by ophthalmologists. The diagnosis is often delayed and difficult. In one study describing three children with this disease, the interval between onset of symptoms and diagnosis was 11, 10 and 10 years. Knowledge of the typical lid lesions considered pathognomonic for the disease may help to diagnose many atypical cases.

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REFERENCES

1. Black MM, Gawkrödger DJ, Seymour CA, Weismann K. Metabolic and nutritional disorders. In: Champion RH, Burton JL, Burns DA, Breathnach SM, eds. *Rook/Wilkinson/Ebling Textbook of Dermatology*, Vol. 2, 6th edn. Oxford: Blackwell Science, 1998; 2640–42.

2. Navarro C, Fachal C, Rodriguez C, Padro L, Dominguez C. Lipoid proteinosis. A biochemical and ultrastructural investigation of two new cases. *Br J Dermatol* 1999; **141**: 362–31.
3. Rosenthal AR, Duke JR. Lipoid proteinosis: case report of direct lineal transmission. *Am J Ophthalmol* 1967; **64**: 1120–25.
4. Lapiere CM. Lipoid proteinosis. In: Freedberg IM, Eisen AZ, Wolff K *et al.*, eds. *Fitzpatrick's Dermatology in General Medicine*, Vol. 2, 5th edn. New York: McGraw-Hill, 1999; 1825–8.
5. Bozdag KE, Gul Y, Karaman A. Lipoid proteinosis. *Int J Dermatol* 2000; **39**: 203–4.
6. Harper JI, Duance VC, Sims TJ, Light ND. Lipoid proteinosis: an inherited disorder of collagen metabolism. *Br J Dermatol* 1985; **113**: 145–51.
7. Fransway AF, Muller SA. Neurocutaneous diseases. In: Moschell SL, Hurley HJ, eds. *Dermatology*, 3rd edn. Philadelphia: WB Saunders 1992; 2054.
8. Staut CCV, Naidich TP. Urbach–Weithe disease (lipoid proteinosis). *Pediatr Neurosurg* 1998; **28**: 212–14.
9. Thappa DM, Gupta S. Images in clinical practice. Eyelid beading – a useful diagnostic clue for lipoid proteinosis. *Indian Pediatr* 2001; **38**: 97.
10. Bohme M, Wahlgren CF. Lipoid proteinosis in three children. *Acta Paediatr* 1996; **85**: 1003–5.

Intraorbital ophthalmic artery aneurysms

ABSTRACT

Angiographically documented aneurysms of the intraorbital ophthalmic artery are extremely rare. Two cases are presented of intraorbital ophthalmic artery aneurysms, both associated with arteriovenous malformation (AVM). There have only been isolated reports documenting this combination of vascular anomalies. In the first case, the aneurysm was treated conservatively and in the second case, after staging angiography and interventional embolization, the aneurysm was removed along with the associated AVM. The clinical presentation, pathogenesis and management of this rare combination of vascular anomalies is briefly discussed.

Key words: cerebral arteriovenous malformation, intraorbital ophthalmic artery aneurysm.

INTRODUCTION

Angiographically documented aneurysms of the intraorbital ophthalmic artery are extremely rare. We present two cases of intraorbital ophthalmic artery aneurysms, both associated with an arteriovenous malformation (AVM).

CASE REPORTS

Case 1

A 47-year-old epileptic man was found unconscious 18 months prior to his referral, presumably postictal. Computed tomographic (CT) scanning incidentally demonstrated a dural AVM located in the anterior cranial fossa. Subsequent angiography revealed a focal left anterior cranial fossa floor dural AVM, fed by ethmoid branches



Figure 1. Cerebral angiogram, via the right innominate artery, lateral view, demonstrating a 4-mm aneurysm (large arrow) at the junction of the first and second parts of the intraorbital ophthalmic artery.

of both ophthalmic arteries. It was drained by a left parasagittal vein and also by a smaller dural base vein. The right ophthalmic artery displayed a 4-mm, saccular aneurysm at the junction of the first and second parts of the intraorbital ophthalmic artery (Fig. 1) Comprehensive ophthalmic examination was normal.

Because of the risk of visual loss with intervention, conservative management of the ophthalmic artery aneurysm was recommended. A similar conservative approach was suggested regarding his dural AVM. The patient has subsequently remained asymptomatic.

Case 2

A 44-year-old man had developed a large, vascular lesion of the left orbit after a kick to the head aged 15. He underwent several debulking operations in the teenage years. After a period of quiescence, the lesion increased in size over the 18 months prior to referral. The lesion caused a chronic ache and minor epiphora. Recently, the patient had become aware of a decrease in vision in the left eye.

Unaided visual acuity was 6/5 on the right, 6/36 on the left, correcting to 6/9. A large, soft, pulsatile vascular lesion was noted overlying the left superomedial orbit. Hertel exophthalmometry was 23 mm on the right, 29 mm on the left. Anterior segment examination was normal. Intraocular pressure was 15 mmHg on the right and 27 mmHg on the left. Fundal examination was unremarkable. Ishihara plate testing was 14/15 for both eyes.

Magnetic resonance imaging confirmed the presence of a predominantly extraconal, left anterior orbital AVM supplied via the left ophthalmic artery. Angiography displayed an enlarged 3 mm left ophthalmic artery, with several branches feeding the AVM. A 6-mm aneurysm was present on the proximal nidus, with at least two smaller aneurysms noted (Fig. 2). Venous drainage was mainly



Figure 2. Angiogram of the left internal carotid artery, lateral view, displaying two small proximal aneurysms (small arrows) and a 6-mm distal aneurysm (large arrow) located adjacent to the orbital arteriovenous malformation.

via the facial and scalp vessels. There was no significant arteriovenous fistula component to the lesion. The patient requested surgical intervention to relieve chronic pain and improve cosmesis.

Prior to surgical resection of the AVM, embolization utilizing Bernstein liquid coils was performed. Three coils were successfully embolized into the distal ophthalmic artery, including the largest aneurysm. However, during injection of a fourth coil, a more proximal coil was dislodged, prolapsing into the internal carotid artery, lodging and partially occluding the posterior temporal branch of the middle cerebral artery. The patient was anticoagulated on heparin. The only neurological deficit noted was transient paraesthesia of the right arm. This resolved completely within 7 days.

Two weeks later the patient underwent complete excision of the AVM via a vertical lid-split approach. The AVM and closely associated aneurysm were meticulously dissected with difficulty via an en bloc technique. Bipolar cautery to the numerous connecting vessels was extensively used to control haemorrhage. The patient was discharged on low dose aspirin. Two months postoperatively the patient was asymptomatic with a corrected visual acuity of 6/6. Further revision surgery to the excess upper eyelid tissue is planned.

DISCUSSION

The aetiology, clinical features and natural progression of intra-orbital ophthalmic artery aneurysms are not completely understood. Presentation is varied, ranging from progressive central field loss, decreased visual acuity, proptosis, diplopia to optic atrophy.^{1–5} The expanding mass of the aneurysm is believed responsible for most symptoms and signs. The aneurysm may rarely rupture, with one case of spontaneous rupture and another secondary to trauma being documented.^{6,7} Both cases resulted in immediate and complete loss of vision. There may be no specific ocular symptoms related to the aneurysms, being detected incidentally during unrelated radiological investigation³ as in our first case. Aneurysms usually arise on either the first^{1,2,4,8} or second^{3,6,7} parts of the intraorbital ophthalmic artery.

Cerebral vascular aneurysm and AVM may coexist.⁹ The incidence of association remains controversial, ranging from 2.7 to 9.3%.^{10,11} Aneurysms specifically associated with dural AVM^{3,12} or traumatic AVM of the eyelid or orbit are extremely rare.¹³ Three theories have been suggested to explain the pathogenesis of such associations:⁹ (i) both are coincidental; (ii) both are the result of multiple congenital vascular anomalies; or (iii) disordered blood flow in the arterial feeders of the AVM predisposes to aneurysmal development. The third theory, relating to haemodynamic factors is favoured in the literature^{3,9,12} and probably was the mechanism in both our cases. Support for this argument is gained by reports of shrinkage of aneurysms on vessels feeding cerebral AVM following surgical resection of the malformation.^{9,10,14} However, it is plausible that a pre-existing arterial wall weakness, congenital or acquired, would predispose to the development of an aneurysm in the abnormal haemodynamic environment surrounding an AVM.

Management guidelines are evolving. Previous series regarding cerebral aneurysms associated with AVM indicate a high mortality rate with conservative management.¹⁵ However, there is a high risk of site-specific complications, particularly blindness, with any intervention of the intraorbital ophthalmic artery. We would generally recommend cautious observation of intraorbital aneurysms, as in case 1. Intervention could be considered if the patient has significant and progressive symptoms from either the aneurysm or the associated AVM. In case 2, the aneurysm was removed fortuitously due to its intimate anatomical association with the AVM.

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REFERENCES

- Ogawa A, Tominaga T, Yoshimoto T, Kiyosawa M. Intra-orbital ophthalmic artery aneurysm. *Neurosurgery* 1992; **31**: 1102–4.
- Rubinstein M, Wilson G, Levin D. Intraorbital aneurysms of the ophthalmic artery. *Arch Ophthalmol* 1968; **80**: 42–4.
- Kikuchi K, Kowada M. Case report: saccular aneurysm of the intraorbital ophthalmic artery. *Br J Radiol* 1994; **67**: 1134–5.
- Raitta C. Ophthalmic artery aneurysm causing optic atrophy and enlargement of the optic foramen. *Br J Ophthalmol* 1968; **52**: 707–9.
- Rengachary S, Kishore P. Intraorbital ophthalmic aneurysms and arteriovenous fistulae. *Surg Neurol* 1978; **9**: 35–41.
- Meyerson L, Lazar S. Intraorbital aneurysm of the ophthalmic artery. *Br J Ophthalmol* 1971; **55**: 199–204.
- Rahmat H, Abbassioun K, Amirjamshidi A. Pulsating unilateral exophthalmos due to traumatic aneurysm of the intraorbital ophthalmic artery. *J Neurosurgery* 1984; **60**: 630–2.
- Jain K. Saccular aneurysm of the ophthalmic artery. *Am J Ophthalmol* 1970; **69**: 997–8.
- Higashi K, Hatano M, Yamashita T, Inoue S, Matsumura T. Coexistence of posterior inferior cerebellar artery aneurysm and arteriovenous malformation fed by the same artery. *Surg Neurol* 1979; **12**: 405–8.
- Miyasaka K, Wolpert S, Prager R. The association of cerebral aneurysms, infundibula, and intracranial arteriovenous malformations. *Stroke* 1982; **13**: 196–203.

11. Koulouris S, Rizzoli H. Coexisting intracranial aneurysm and arteriovenous malformation: case report. *Neurosurgery* 1981; 8: 219–22.
12. Kaech D, Tribolet N, Lasjaunias P. Anterior inferior cerebellar artery aneurysm, carotid bifurcation aneurysm, and dural arteriovenous malformation of the tentorium in the same patient. *Neurosurgery* 1987; 21: 575–81.
13. Holt G, Holt J, Cortez E, Thornton W, Young W. Traumatic facial arteriovenous malformations. *Laryngoscope* 1980; 90: 2011–20.
14. Shenkin H, Jenkins F, Kim K. Arteriovenous anomaly of the brain associated with cerebral aneurysm: case report. *J Neurosurgery* 1971; 34: 225–8.
15. Liu Y, Zhu S, Jiao L, Wang H, Li X, Li G. Cerebral arteriovenous malformations associated with aneurysms – a report of 10 cases and literature review. *J Clin Neuroscience* 2000; 7: 254–6.

Optical coherence tomography of adult-onset foveomacular vitelliform dystrophy

ABSTRACT

Adult-onset foveomacular vitelliform dystrophy is characterized by a focal, round or oval shaped, subretinal foveal yellow lesion that presents bilaterally and usually symmetrically in the fourth or fifth decade of life. In addition to clinical observation, diagnosis of the disease process has until now been assisted by electrophysiological and fundus fluorescein studies. Two cases are presented that correlate the clinical fundus appearance with optical coherence tomography findings and describe the location of the yellow vitelliform material.

Key words: adult-onset foveomacular vitelliform dystrophy, adult vitelliform macular degeneration, pseudovitelliform macular degeneration, Best's disease.

INTRODUCTION

Adult-onset foveomacular vitelliform dystrophy (AOFVD) comprises one of the 'Pattern' dystrophies of the retinal pigment epithelium (RPE). It is also known as adult-onset foveomacular pigment epithelial dystrophy,¹ adult vitelliform macular degeneration,^{2,3} and pseudovitelliform macular degeneration.^{4,5} The classic presentation of the disease is a bilateral and symmetrical maculopathy where the primary lesions are composed of yellow deposits of abnormal lipofuscin. Histological studies vary in their interpretation of the location of the vitelliform material.^{6–9} Patients usually present in the fourth or fifth decade of life with round or oval one-third disc diameter yellow subfoveal lesions within the macular region. Vision is initially normal, or only slightly reduced, and mild metamorphopsia may be noted.

The classification of this disorder as a dystrophy or a degeneration has resulted in some debate. It is generally regarded as autosomal dominant, with diminished penetrance resulting in highly variable phenotypic expressions.¹⁰ The bilaterality, symmetry and frequent positive family involvement suggests a dystrophy, while the late

onset and similarity, both clinically and histopathologically, to age-related macular degeneration is suggestive of a degeneration. Electrophysiological (electro-oculogram) studies are typically normal.

Optical coherence tomography (OCT) is a relatively new technique for non-invasive, cross-sectional visualization of the retina. It can provide objective and quantitative measurements of retinal structure, similar to that obtained from histological sections.¹¹ OCT is analogous to ultrasonography, except that an optical rather than acoustic signal enables a higher image resolution in the retina. The longitudinal resolution of 10 µm capable with OCT is significantly better than other non-invasive imaging techniques of the eye, including standard (150 µm)¹² and high frequency (20 µm)¹³ ultrasound. Ultrasound is not suitable for the posterior segment of the eye because of poor signal penetration. The resolution of computed tomography and magnetic resonance imaging^{14,15} is limited to hundreds of microns and cross-sectional imaging of the fundus with scanning laser ophthalmoscopy¹⁶ and scanning laser tomography¹⁷ are restricted by pupil size and ocular aberrations to about 300 µm.

The commercially available OCT instrument operates on the principle of partial coherence interferometry. It scans an 820-nm infrared light beam from a superluminescent diode across the retina and generates a cross-sectional image of the tissue by recording the scattering profile versus depth of each transverse location of the beam. The scattering profile is measured by interfering the back-scattered light from the tissue with reference light from a variable length arm in a fibre optic Michelson interferometer. The resulting tomograph that is displayed is a series of 512 one-dimensional longitudinal measures of reflectivity versus depth (analogous to an A-Scan ultrasound). Retinal thickness and retinal nerve fibre layer thickness are calculated through image processing of the resulting cross-sectional OCT image. The algorithm detects these boundaries by searching each A-scan axially for the highest rates of change in reflectivity.

The third generation OCT (OCT 3) has recently been released (Zeiss, Germany). The previous OCT 2 technology has considerable limitations in comparison to OCT 3. Perhaps the most important feature of OCT 3 is that it has twice the depth (longitudinal) resolution of OCT 2, being able to resolve to 10 µm in tissue as opposed to 20 µm. This allows for far superior visualization and subsequent interpretation of intraretinal (and deeper) structures. Figure 1 shows the difference in resolution capability between OCT 2 and OCT 3 of a foveal image. In processing the image information, OCT 3 also utilizes 10-fold the datapoints of OCT 2 (OCT 3 scan datapoints = 524 288 vs 50 000 OCT 2).

Ultrahigh-resolution OCT appears to be the next step in the OCT evolutionary ladder. This technology was developed for use in a study by Drexler *et al.*¹⁸ and utilizes a titanium-sapphire laser to provide an axial resolution capability of 3 µm. It provides superior imaging of intraretinal structures, such as the ganglion cell layer, the photoreceptor layer and the RPE, compared to OCT 3 technology.

We report two cases of AOFVD and in doing so correlate the clinical findings with imaging obtained through OCT.

CASE REPORTS

Case 1

A 79-year-old man presented for routine review in January 2002. He had been diagnosed with glaucoma in 1986 and was currently medicated with betaxolol 0.25% b.i.d. OU. He felt that his vision

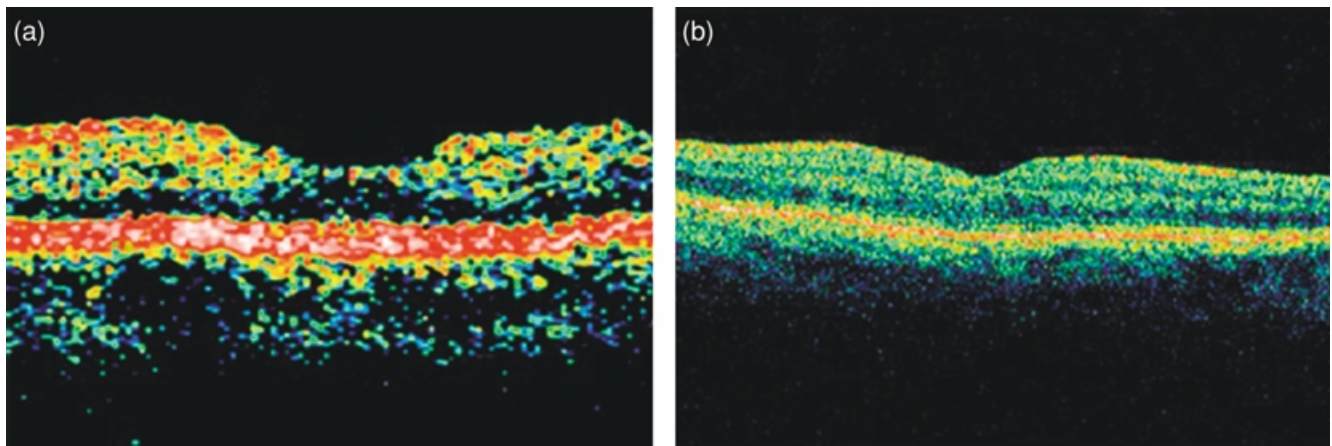


Figure 1. Optical coherence tomography (OCT) resolution capability for foveal images with (a) OCT 2 and (b) OCT 3.

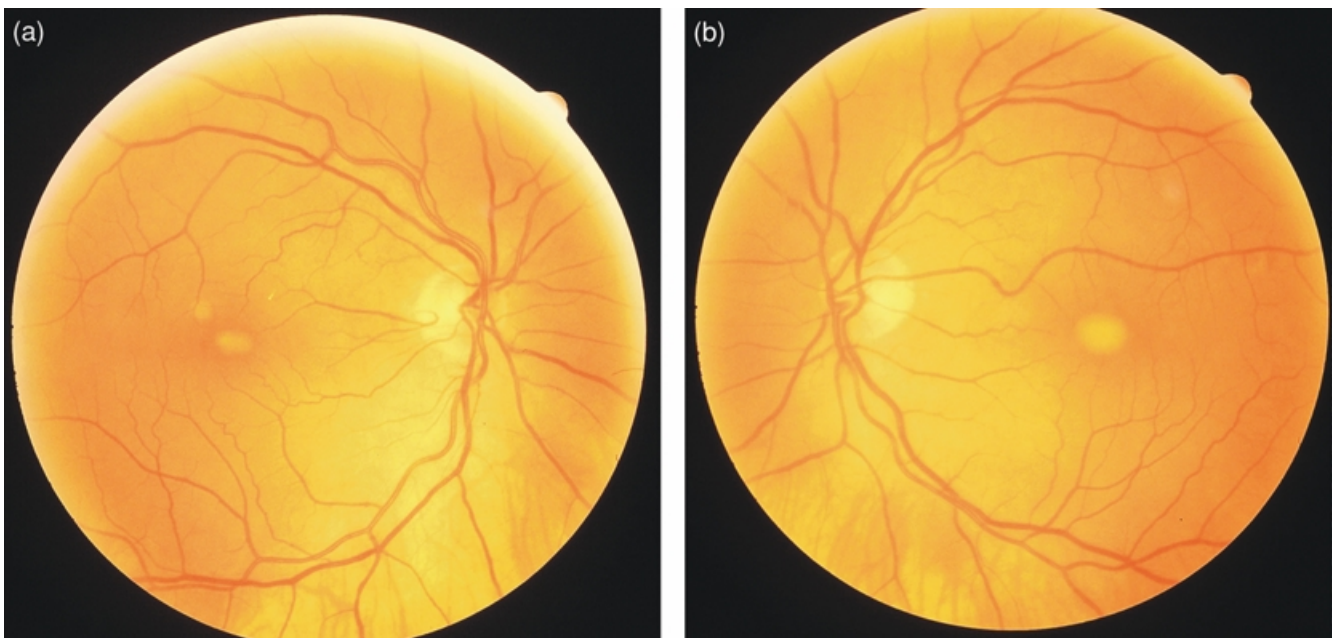


Figure 2. Fundus photographs showing vitelliform lesions of (a) right eye and (b) left eye of patient in case 1.

had been becoming increasingly blurred over the past 12 months and his left eye was markedly worse. He was taking antihypertensive medication. Visual acuities were measured as right 6/9 and left 6/18. Intraocular pressures were right 15 mmHg and left 19 mmHg. Mild nuclear sclerotic cataracts (2+) were noted in both eyes. The glaucoma appeared stable with controlled intraocular pressure and cup/disc ratios of 0.5 right and left. Both maculae showed a discrete, oval, yellow lesion with the left lesion appearing larger (Fig. 2).

At review 12 months later, pigmentary changes were recorded at the level of the RPE at both maculae and a diagnosis of a pattern dystrophy was made. Ophthalmoscopically, the lesions had changed in appearance and a new diagnosis of adult vitelliform degeneration formed. Fluorescein angiography was not performed as the patient reported a previous systemic reaction to the contrast dye. It was felt the risk of performing fluorescein angiography could not be justified based on the ophthalmoscopically secure diagnosis.

Optical coherence tomography was performed. This revealed a significant thickening and distortion of the foveal architecture in both eyes (left > right). Figure 3 illustrates the specific location of the intraretinal structures on the monochromatic OCT map for the left eye of this patient. Of particular note is the differentiation of the two bright band layers, the junction of the inner and outer photoreceptor segments and the RPE/choriocapillaris complex. Bruch's membrane cannot be resolved within this complex. These layers are identifiable because of variations in tissue reflectivity, which the OCT 3 displays as corresponding bright–dark–bright layers on the monochromatic map. There is a thickening of the foveal RPE with subsequent loss of the foveal depression. The vitelliform lesion is shown as a hyper-reflective structure lying at the level of, or above the retinal pigment epithelium. Figure 4 shows the same scan of the left eye in colour (Fig. 4a) and a scan of the right eye (Fig. 4b) with 90° and 30° scan orientations, respectively.

Case 2

A 66-year-old woman was referred for assessment and consideration of repair of a macular hole in the right eye. Her general health was good apart from hypertension and osteoporosis. In 1998, visual acuities were recorded as right 6/6 and left 6/12. Ophthalmoscopically, bilateral foveal lesions were noted, consistent with a diagnosis of adult vitelliform macular dystrophy. At her most recent review, a stage IV macular hole had developed in the right eye (Fig. 5a) and it was our feeling that the progressive loss of normal foveal architecture gave rise to this sequelae. The foveal lesion in the left eye was still consistent with a diagnosis of vitelliform dystrophy (Fig. 5b). Fluorescein angiography revealed a typical adult vitelliform lesion in the left eye with early blockage and late hyperfluorescence (Fig. 6). No choroidal neovascular membrane was observed in either eye. Visual acuities were measured as right 6/18 and left 6/18. Figure 7 shows OCT tomographs for the right and left eyes, respectively.

DISCUSSION

Optical coherence tomography is a new intraretinal imaging modality that has been shown to be useful in providing diagnostic

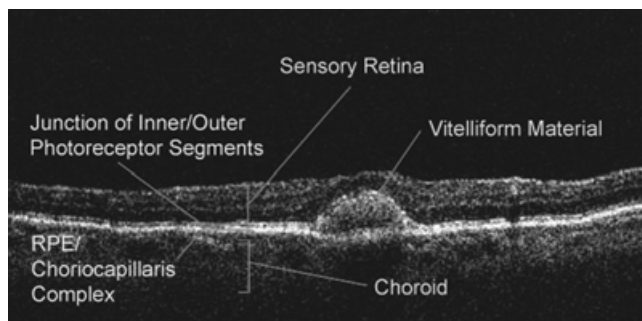


Figure 3. Monochromatic optical coherence tomography image of left eye (case 1) correlating retinal/choroidal anatomical structures.

information about various macular diseases.^{19,20} The OCT 3 tomographs presented in these case reports had an acquisition time of 1 s each. The images are displayed in false colour, where bright colours (red to white) correspond to regions of high relative optical reflectivity or back-scattering, and dim colours (blue to black) represent areas of minimal or no relative reflectivity. Thus, as a basic principle of OCT, two adjacent structures can be differentiated from each other if their reflectivity is different. Interpretation of OCT tomographs requires some knowledge of the correlation between healthy retinal anatomy and features evident on OCT.

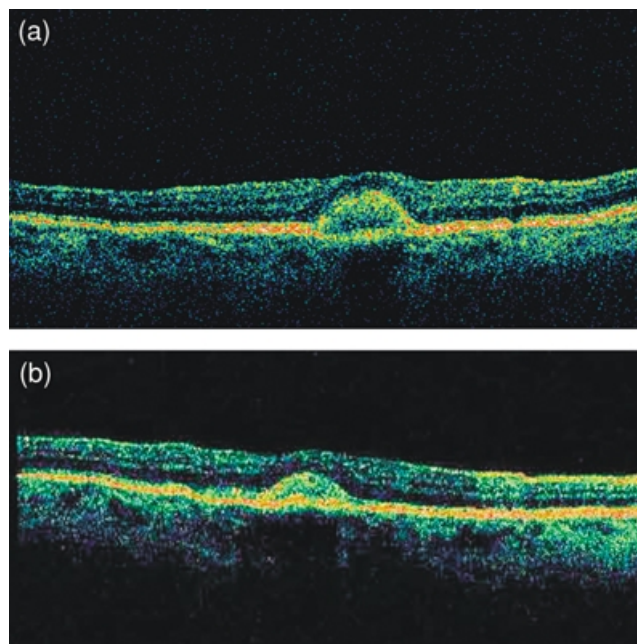


Figure 4. Optical coherence tomography images of (a) left eye and (b) right eye of patient in case 1. The top image represents a 90° (vertical) optical section through the fovea and the bottom image is at a 30° orientation (anticlockwise convention).

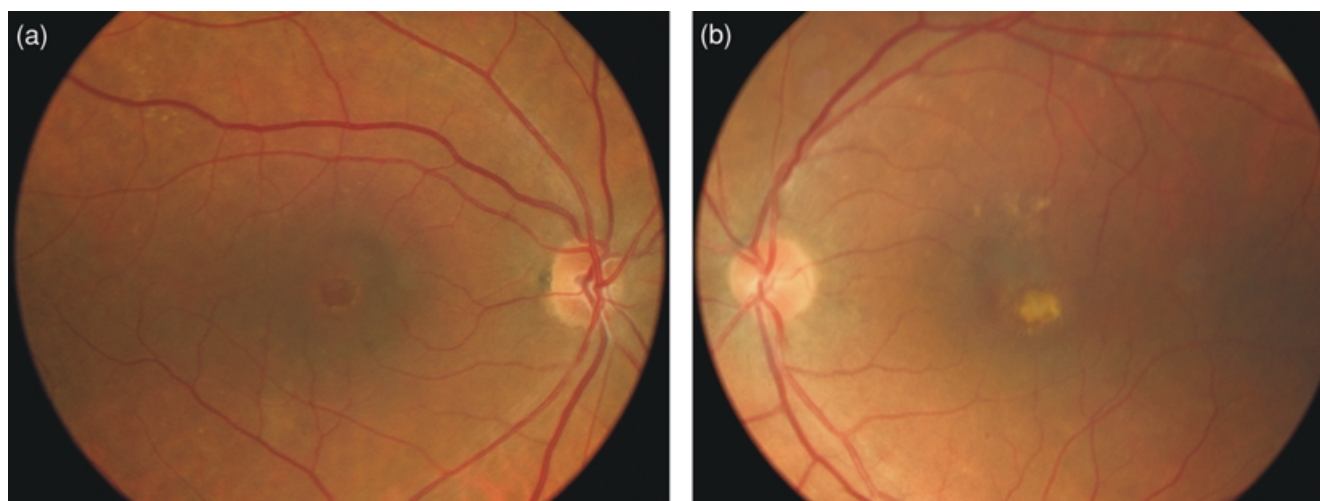


Figure 5. Fundus photographs showing vitelliform lesions of (a) right eye and (b) left eye of patient in case 2.

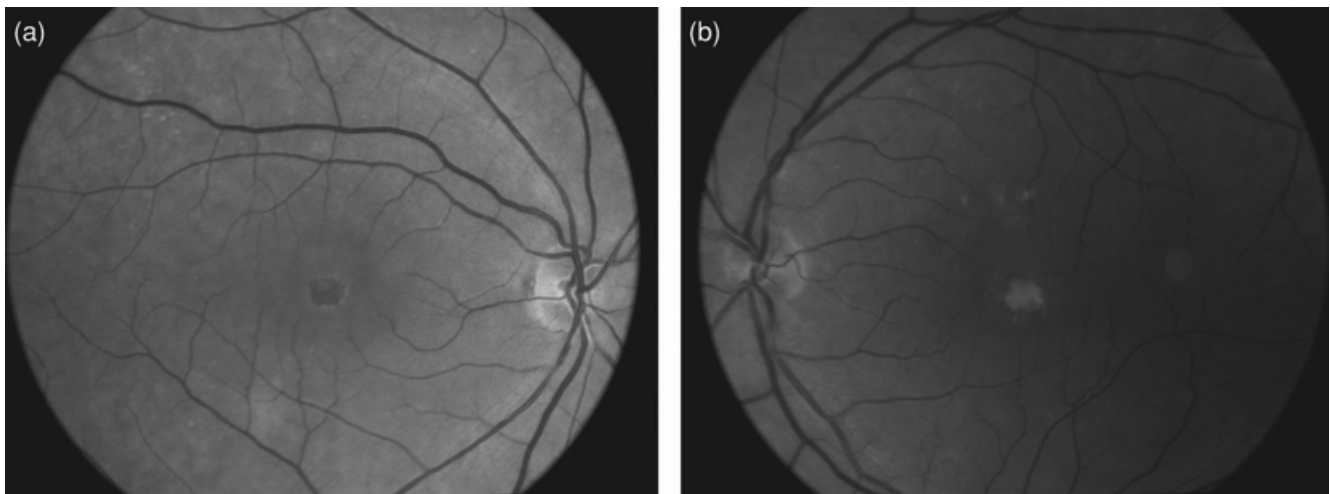


Figure 6. Fluorescein angiographs showing vitelliform lesions of (a) right eye and (b) left eye of patient in case 2.

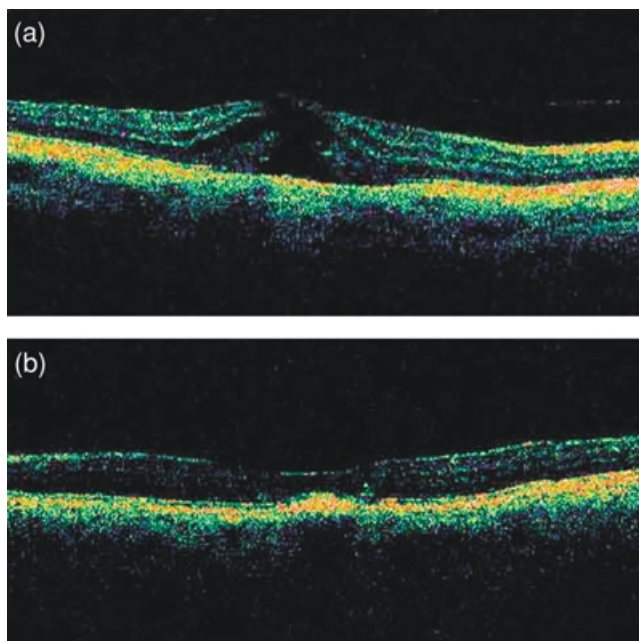


Figure 7. Optical coherence tomography images of (a) right eye and (b) left eye of patient in case 2. Both optical sections are at a 120° orientation (anticlockwise convention).

The vitreoretinal interface is defined by the contrast between the non-reflective vitreous and the back-scattering surface of the retina. The fovea is clearly identifiable due to its characteristic morphology, including the foveal depression and the lateral displacement of the retina anterior to the photoreceptors and Henle's fibre layer.

Unlike other retinal imaging techniques and earlier OCT technology, OCT 3 differentiates between light back-scattered from different depths in the retina, permitting visualization of retinal layer structure. The intermediate layers of the sensory retina exhibit moderate back-scattering of varying reflectivity corresponding to different retinal layer densities. The inner margin of the retina is

revealed by a bright back-scattering red layer that corresponds in location to the retinal nerve fibre layer. The next anatomical layer of distinction is a thin red band of high reflectance immediately superior to the RPE/choriocapillaris complex. This represents the interface between inner and outer photoreceptor segments and is not yet fully understood, although it has been identified by correlating ultrahigh-resolution OCT images with histologic features.²¹ Below this photoreceptor junction, another highly reflective broader red layer (approximately 70 µm thick) delineates the posterior boundary of the retina in the tomograph and corresponds to the RPE and choriocapillaris. Beneath the choriocapillaris, relatively weak reflections are observed from the deep choroid and sclera, due to attenuation of the signal after passing through the retina.²²

Optical coherence tomography imaging of AOFVD was first demonstrated by Puliafito *et al.* in 1996.²³ Because there have been only a few histopathological cases of AOFVD reported, there has been uncertainty in the scientific literature as to the specific location of the vitelliform material. The exact location of the material, whether below, above or inside the RPE, has not yet been conclusively determined, either by clinical or histological assessment. In one recent report utilizing OCT, the AOFVD material was described as a focal detachment of the RPE.²⁴ Other reports basing their findings on analysis of stereoscopic fundus photographs have specified the location of the material as subretinal,²⁵ at the level of,²⁶ or under the RPE.²⁷ Histopathological studies are supportive of a sub-RPE location.^{8,28} In contrast, another recent study of AOFVD using OCT by Benhamou *et al.* describes the location as 'under the sensory retina, but above the retinal pigment epithelium'.²⁹ Our findings in the two cases presented suggest that the vitelliform lesion is located at the level of, or above the RPE. In both instances there is a focal elevation of the neurosensory retina caused by the vitelliform material with loss of the normal foveal depression. One cannot be certain of the exact location because of the inability of OCT 3 to resolve and differentiate the RPE, Bruch's membrane and choriocapillaris.

Current OCT 3 provides more detailed structural information than any other ophthalmic diagnostic technique. However, many early pathologic retinal changes including those associated with AOFVD are still below the resolution limit of OCT 3. Drexler *et al.*

used a recently developed ultrahigh-resolution OCT to demonstrate imaging of intraretinal microstructures and visualize intraretinal morphology of macular pathology.¹⁸ The major intraretinal layers were visualized non-invasively *in vivo*, and images had excellent correlation with known retinal morphologic features. Their OCT findings of AOFVD describe 'a well-circumscribed elevation of the sensory retina above the thickened RPE'.

In the second case report, a full thickness macular hole developed in the patient's right eye. The hypothesis for the aetiology of idiopathic macular hole is tangential vitreous traction. By contrast, foveal thinning and atrophy is considered to be the main pathophysiological trigger in AOFVD.³⁰

Optical coherence tomography is a useful direct cross-sectional imaging technique that may be helpful in the early diagnosis and more sensitive monitoring of a range of macular and retinal diseases. These case reports illustrate that AOFVD has characteristic OCT findings that allow simple, rapid, non-invasive diagnosis of this infrequent macular disorder.

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REFERENCES

- Vine AK, Schatz H. Adult-onset foveomacular pigment epithelial dystrophy. *Am J Ophthalmol* 1980; **89**: 680–91.
- Bloom LH, Swanson DE, Bird AC. Adult vitelliform macular degeneration. *Br J Ophthalmol* 1981; **65**: 700–1.
- Greaves AH, Sarks JP, Sarks SH. Adult vitelliform macular degeneration: a clinical spectrum. *Aust NZ J Ophthalmol* 1990; **18**: 171–8.
- Fishman GA, Trimble S, Raab MF, Fishman M. Pseudovitelliform macular degeneration. *Arch Ophthalmol* 1977; **95**: 73–6.
- Hodes BL, Feiner LA, Sherman SH, Cunningham D. Progression of pseudovitelliform macular dystrophy. *Arch Ophthalmol* 1984; **102**: 381–3.
- Jaffe GJ, Schatz H. Histopathologic features of adult-onset foveomacular pigment epithelial dystrophy. *Arch Ophthalmol* 1988; **106**: 958–60.
- Gass JDM. A clinicopathological study of a peculiar foveomacular dystrophy. *Trans Am Ophthalmol Soc* 1974; **72**: 139–56.
- Patrinley JR, Lewis RA, Font RL. Foveomacular vitelliform dystrophy, adult type; a clinicopathologic study including electron microscopic observations. *Ophthalmology* 1985; **92**: 1712–8.
- Dubovy S, Hairston R, Schatz H, Schachat A *et al.* Adult-onset foveomacular pigment epithelial dystrophy: clinicopathologic correlation of three cases. *Retina* 2000; **20**: 638–43.
- Deutman AF. Electro-oculography in families with vitelliform dystrophy of the fovea: detection of the carrier state. *Arch Ophthalmol* 1969; **81**: 305–16.
- Huang D, Swanson EA, Lin CP *et al.* Optical coherence tomography. *Science* 1991; **254**: 1178–81.
- Olsen T. The accuracy of ultrasonic determination of axial length in pseudophakic eyes. *Acta Ophthalmol* 1989; **67**: 141–4.
- Pavlin CJ, Sherar MD, Foster FS. Subsurface ultrasound microscopic imaging of the intact eye. *Ophthalmology* 1990; **97**: 244–50.
- Chang DF. Ophthalmic examination. In: Vaughan DG, Asbury T, Riordan-Eva P, eds. *General Ophthalmology*, 13th edn. East Norwalk: Appleton & Lange, 1992; 30–62.
- Seiler T, Bende T. Magnetic resonance imaging of the eye and orbit. In: Masters BR, eds. *Noninvasive Diagnostic Techniques in Ophthalmology*. New York: Springer-Verlag, 1990; 17–31.
- Webb RH, Hughes GW, Delori FC. Confocal scanning laser ophthalmoscope. *Appl Opt* 1987; **26**: 1492–9.
- Billie JF, Dreher GW, Zinser G. Scanning laser tomography of the living human eye. In: Masters BR, eds. *Noninvasive Diagnostic Techniques in Ophthalmology*. New York: Springer-Verlag, 1990; 528–47.
- Drexler W, Sattmann H, Hermann B, Ko T *et al.* Enhanced visualisation of macular pathology with the use of ultra-high resolution optical coherence tomography. *Arch Ophthalmol* 2003; **121**: 695–707.
- Coker JC, Duker JS. Macular disease and optical coherence tomography. *Curr Opin Ophthalmol* 1996; **7**: 33–8.
- Iida T, Hagimura N, Sato T *et al.* Evaluation of central serous chorioretinopathy with optical coherence tomography. *Am J Ophthalmol* 2000; **129**: 16–20.
- Gloesmann M, Hermann B, Schubert C *et al.* Histologic correlation of pig retina radial stratification with ultrahigh-resolution optical coherence tomography. *Invest Ophthalmol Vis Sci* 2003; **44**: 1696–702.
- Hee MR, Izatt JA, Swanson EA *et al.* Optical coherence tomography of the human retina. *Arch Ophthalmol* 1995; **113**: 325–32.
- Puliafito CA, Hee MR, Schuman JS *et al.* *Optical Coherence Tomography of Ocular Disease*. Thorofare: Slack, 1996; 269–73.
- Yamaguchi K, Yoshida M, Kano T, Itabashi T, Yoshioka Y, Tamai M. Adult onset foveomacular vitelliform dystrophy with retinal folds. *Jpn J Ophthalmol* 2001; **45**: 533–4.
- Lim JI, Enger C, Fine SL. Foveomacular dystrophy. *Am J Ophthalmol* 1994; **117**: 1–6.
- Sabates R, Pruett RC, Hirose T. Pseudovitelliform macular degeneration. *Retina* 1982; **2**: 197–205.
- Snyder DA, Fishman GA, Witteman G, Fishman M. Vitelliform lesions associated with retinal pigment epithelial detachment. *Ann Ophthalmol* 1978; **10**: 1711–15.
- Weingest TA, Kobrin JL, Watzke RC. Histopathology of Best's macular dystrophy. *Arch Ophthalmol* 1982; **100**: 1108–14.
- Benhamou N, Souied E, Zolf R, Coscas F, Coscas G, Soubrane G. Adult-onset foveomacular vitelliform dystrophy: a study by optical coherence tomography. *Am J Ophthalmol* 2003; **135**: 362–7.
- Noble KG, Chang S. Adult vitelliform degeneration progressing to full-thickness macular hole. *Arch Ophthalmol* 1991; **109**: 325.